Section 18 System level

Contents

1. INTRODUCTION

This system level section is divided into two parts:

- Fine adjustments of the system.
- Safety actions.

2. SYSTEM RESONANCE FREQUENCY DETERMINATION

The adjustment of the magnet field to the desired F0 frequency is done in the magnet section.
To enter the actual F0 frequency into the HW parameter file, the following test must be performed:

Pre-conditions.

The following procedures must have been performed (see section 14):

- The "automatic BDAS procedures".
- Max KW adjustment .

F0 determination.

1. Login : GYROSCAN.
2. Select: SCAN CONTROL.
3. Select: SCAN UTILITIES.
4. Select: ENTER SERVICE MODE
5. Select: SYSTEM TUNING
6. Select: SYSTEM TUNING TOOLS.
7. Select: INSTALLATION PROCEDURES.
8. Select: F0 DETERMINATION
9. Follow the instructions.
   Start the measurement.
   If the frequency is within specification no adjustment is necessary.

Specifications:

Value for Intera 0.5T:
Resonance frequency at system installation is: 21.295 MHz ± 5 kHz

Value for Intera 1.0T:
Resonance frequency at system installation is: 42.595 MHz ± 10 kHz

Value for Intera 1.5T:
Resonance frequency at system installation is: 63.897 MHz ±15 kHz

10. Select: EXIT

3. RF POWER REFERENCE AND PU COIL MEASUREMENT

3.1. PRE-CONDITIONS

- The "automatic BDAS procedures" must have been executed.
- The "RF amplifier adjustments" must have been executed.
- Resonance Frequency Determination must have been done.
- The RF door must be closed.

3.2. PROCEDURE FOR RF POWER REFERENCE AND PU COIL MEASUREMENT

3.2.1. POWER REFERENCE MEASUREMENT

1. Login : GYROSCAN.
2. Select: SCAN CONTROL.
3. Select: SCAN UTILITIES.
4. Select: ENTER SERVICE MODE
5. Select: SYSTEM TUNING
6. Select: SYSTEM TUNING TOOLS.
7. Select: INSTALLATION PROCEDURES.
8. Select: RF POWER REF. AND PU COIL MEASUREMENTS.
9. Select: RF POWER REFERENCE MEASUREMENT QBC.
10. Follow the instructions on the screen
11. Select: <Proceed> to start the measurement.

NOTE

The Power Reference measurement and Pickup triplevel measurement are influenced by the position of the pickup coils. In these measurements errors will be generated when the positioning is not in specification. The Pickup Cal B1 is the sensitivity [uT/V] of the Pickup coils measured with a known B1 field. With the measured voltage the sensitivity is calculated. When the found pickup Cal B1 value is too low, the pickup coils are too close to the QBC and should be moved further away from the QBC. Use 1 mm step at the time for both pickup coils. When the found pickup value is too high the pickup coils should be moved towards the QBC.

3.2.2. PICKUP-COIL MEASUREMENTS

1. Login : GYROSCAN.
2. Select: SCAN CONTROL.
3. Select: SCAN UTILITIES.
4. Select: ENTER SERVICE MODE.
5. Select: SYSTEM TUNING.
6. Select: SYSTEM TUNING TOOLS.
7. Select: INSTALLATION PROCEDURES.
8. Select: RF POWER REF. AND PU COIL MEASUREMENTS.
9. Select: PICKUP COIL TRIPLEVEL ADJUSTMENT.
10. Carefully read the instructions on the screen. Upgraded systems may have to connect the 50E dummy load in stead of the 50E load of the body coil depending on their configuration!
11. After reading the instructions, the measurement must be started with <Proceed>.
12. Write down the testresult in the MRL (section 22, Safety checks)

4. GRADIENT SYSTEM

4.1. GRADIENT PARAMETERS DETERMINATION

The gradient parameter determination is necessary for optimizing gradient performance. The Eddy current analysis tool will analyze the linear eddy currents. With the determined parameters the compensation for an EPI – diffusion measurement is calculated.

4.1.1. PROCEDURE FOR GRADIENT PARAMETERS DETERMINATION

Precondition: The gradient cabinet must have been switched on for at least 15 minutes.

1. Login : GYROSCAN.
2. Select: SCAN CONTROL.
3. Select: SCAN UTILITIES.
4. Select: ENTER SERVICE MODE
5. Select: SYSTEM TUNING.
6. Select: SYSTEM TUNING TOOLS.
7. Select: INSTALLATION PROCEDURES.
8. Select: GRADIENT PARAMETERS DETERMINATION.
9. Answer the question : Run all subsequent tools successively [No,Yes]: with Yes
10. Select: <Proceed> to start all three measurements.
    The instructions will appear on the screen.

The next measurements will be performed automatically:
1. DAC linearity calibration
2. Gradient channel delay determination
3. Gradient channel phase adjustment
4. Gradient FID shimming

4.2. EDDY CURRENT ANALYSIS

4.2.1. PROCEDURE FOR EDDY CURRENT ANALYSIS

NOTE

In contrast with other measurements, the position of the 3 litre bottle on the patient table must be vertical for the Eddy current analysis procedure and not horizontal.

1. Select: SCAN UTILITIES.
2. Select: ENTER SERVICE MODE.
3. Select: SYSTEM TUNING.
4. Select: SYSTEM TUNING TOOLS.
5. Select: INSTALLATION PROCEDURES.
6. Select: EDDY CURRENT ANALYSIS
7. Select: <Proceed> to start all three measurements.
   The instructions will appear on the screen.

4.3. FINE ADJUSTMENT GRADIENT STRENGTH

4.3.1. CHECK TUNED HARDWARE PARAMETERS

For a correct procedure, the gradient gain value in the Tuned hardware parameterset must be set to 0.

Procedure:
1. Select. SCAN CONTROL
2. Select· SCAN UTILITIES.
3. Select: ENTER SERVICE MODE
4. Select: SYSTEM TUNING
5. Select: TUNED HARDWARE PARAMETERS

6. Check the present value for the parameters: grad gain adjust X, Y and Z
7. Set the values to 0.
8. Select· STORE to save the changes (a reason must be entered when the parameters were changed.)

4.3.2. MEASUREMENT GRADIENT STRENGTH X AND Y

1. Login : GYROSCAN.
2. Select: SCAN CONTROL
3. Select: SPT SCANNING
4. Select: SERVICE MODE
5. Select: PERFORM IQP/EST MEASUREMENTS
6. Select: ANALYSIS PRINCIPLE: IQP
7. Select: XX MISC (see next page if not in list)
8. Select: Ident GA_XY (scanname: ADJ:XY)
9. Position the head performance phantom with the holder for transversal scans in the head coil. Connect the head coil to one of the surface coil connectors. Move the head coil to the iso-center of the magnet by using the light visor and pressing <Travel to scan-plane>.
10. Start the batch with <PROCEED> (2x)
11. In the batch, SCOUT scans and the scans "GA1:X_grad" and "GA2:Y_grad" are performed automatically.

4.3.3. EVALUATION OF SCANS

1. After the scans have finished select the SPT icon.
2. Select: SERVICE MODE
3. Select: IQ parameter/data evaluation tools
4. Select: spatial linearity measurements
5. Select: Select examination
6. Select: examination GRAD_ADJ
7. Select: Select scan:GA1:X_grad
8. Select: Select image: Slice 1 SE_M
9. Select: Philips quality procedure.
10. Write down the value: Size vertical
    Specification: 149.5 < Size < 150.5 mm.

11. Select: Select scan: GA2:Y_grad
12. Select: Select image: Slice 1 SE_M
13. Select: Philips quality procedure.
14. Write down the value: Size horizontal.
    Specification: 149.5 < Size < 150.5 mm.

4.3.4. MODIFY TUNED HARDWARE PARAMETERS

If both measured values meet the specifications, continue with paragraph 4.4.2 Measurement gradient strength Z. For the gradients that did not meet the specifications, calculate new gradient parameters.

1. Select: SCAN UTILITIES.
2. Select: ENTER SERVICE MODE
3. Select: SYSTEM TUNING
4. Select: TUNED HARDWARE PARAMETERS

The new gradient gain value can be calculated according the next formula (1 mm = 0.66%):

% new = (150 – Measured size) x 0.66%

%new is the new value for the gradient gain adjustment value.

5. Modify the parameters: grad gain adjust X and Y to the new calculated values. (The values must be between - 4.10 % and + 4.09 %)
6. Select. STORE to save the changes (a reason must be entered for the change of parameters)

4.4. VERIFICATION NEW GRADIENT PARAMETERS

1. Login : GYROSCAN.
2. Select: SCAN CONTROL
3. Select: SPT SCANNING
4. Select: SERVICE MODE
5. Select: PERFORM IQP/EST MEASUREMENT
6. Select: ANALYSIS PRINCIPLE: IQP
7. Select: XX MISC (see next page if not in list)
8. Select: Ident GA_XY (scanname ADJ:XY)
9. If not already done: Position the head performance phantom with the holder for transversal scans in the head coil. Connect the head coil to one of the surface coil connectors. Move the head coil to the iso-center of the magnet by using the light visor and pressing <Travel to scan-plane>.
10. Start the batch with <PROCEED> (2x)
11. In the batch, SCOUT scans and the scans "GA1:X_grad" and "GA2.Y_grad" are performed automatically.

4.4.1. VERIFICATION OF GRADIENT STRENGTH

1. After the scans have finished select the SPT icon.
2. Select: SERVICE MODE
3. Select: IQ parameter/data evaluation tools
4. Select: spatial linearity measurements
5. Select: Select examination
6. Select. examination GRAD_ADJ
7. Select: Select scan:GA1:X-grad
8. Select: Select image: Slice 1 SE_M
9. Select: Philips quality procedure.
10. Write down the value: Size vertical
    Specification: 149.5 < Size < 150.5 mm.
11. Select: Select scan: GA2:Y_grad
12. Select: Select image: Slice 1 SE_M
13. Select: Philips quality procedure.
14. Write down the value: Size horizontal.
    Specification: 149.5 < Size < 150.5 mm.
15. Repeat the procedure if not within specification.

4.4.2. MEASUREMENT GRADIENT STRENGTH Z

1. Login : GYROSCAN.
2. Select: SCAN CONTROL
3. Select: SPT SCANNING
4. Select: SERVICE MODE
5. Select: PERFORM IQP/EST MEASUREMENT
6. Select: ANALYSIS PRINCIPLE: IQP
7. Select: XX MISC (see next page if not in list)
8. Select: Ident GA_Z (Scanname ADJ:Z)
9. Position the head performance phantom with the holder for transversal scans in the head coil. Connect the head coil to one of the surface coil connectors. Move the head coil to the iso-center of the magnet by using the light visor and pressing <Travel to scan-plane>.
10. Start the batch with <PROCEED> (2x)
11. In the batch, SCOUT scans and the scan "GA1:Z_grad" will be performed automatically.

4.4.3. EVALUATION OF SCAN

1. After the scans have finished select the SPT icon.
2. Select: SERVICE MODE
3. Select: IQ parameter/data evaluation tools
4. Select: spatial linearity measurements
5. Select: Select examination
6. Select: examination GRAD_ADJ
7. Select: Select scan: GA1:Z_grad
8. Select: Select image slice 3 SE_M
9. Select: Philips quality procedure.
10. Write down the value: Size vertical.
    Specification: 149.5 < Size < 150.5 mm.

4.4.4. MODIFY TUNED HARDWARE PARAMETERS

When the value for the Z-gradient did not meet the specifications, the new gradient parameter must be calculated.

1. Select: SCAN UTILITIES.
2. Select: ENTER SERVICE MODE
3. Select: SYSTEM TUNING
4. Select: TUNED HARDWARE PARAMETERS

The new gradient gain value can be calculated according the next formula (1 mm = 0.66%):

$$\% \text{ new} = (150 - \text{Measured size}) \times 0.66\%$$

%new is the new value for the gradient gain adjustment value.

5. Modify the parameters: grad gain adjust Z to the new calculated values.

(The values must be between - 4.10 % and + 4.09 %)

6. Select: STORE to save the changes (a reason must be entered for the change of parameters)

4.5. VERIFICATION NEW GRADIENT PARAMETERS

1. Login : GYROSCAN
2. Select: SCAN CONTROL
3. Select: SPT SCANNING
4. Select: SERVICE MODE
5. Select: PERFORM IQP/EST MEASUREMENTS
6. Select: ANALYSIS PRINCIPLE. IQP
7. Select: MISC (see next page if not in list)
8. Select: Ident. GA_Z (Scanname: ADJ:Z)
9. Position the head performance phantom with the holder for transversal scans in the head coil. Connect the head coil to one of the surface coil connectors. Move the head coil to the iso-center of the magnet by using the light visor and pressing <Travel to scan-plane>.
10. Start the batch with <PROCEED> (2x)
11. In the batch, SCOUT scans and the scan "GA1:Z_grad" will be performed automatically.

4.5.1. VERIFICATION OF GRADIENT STRENGTH

1. After the scans have finished select the SPT icon.
2. Select: SERVICE MODE
3. Select: IQ parameter/data evaluation tools
4. Select: spatial linearity measurements
5. Select: Select examination
6. Select: examination GRAD_ADJ
7. Select: Select scan: GA1:Z_grad
8. Select: Select image slice 3 SE_M
9. Select: Philips quality procedure.
10. Write down the value: Size vertical.
    Specification: 149.5 < Size < 150.5 mm.
11. Repeat the procedure if not within specification.

5. COMPRESSOR SHUTDOWN FUNCTION TEST

1. Position the Quad head coil with the phantom holder and the head performance phantom in the iso-center.
2. Select a dummy patient
3. Select anatomy
4. Select Phantom studies
5. Select Headt
6. Select Scout
7. Verify that the compressor is running.
8. Start the scan
9. Enter <return> for the default scan name.
10. Verify that the compressor is switched of during the scan
11. Stop the scan with the <stop scan> button.
12. Verify that the compressor start running again.

6. TOTAL BACKUP

To be able to make a fast restore of your software configuration, make a total back-up of the software according the software installation manual.

7. MRL SYSTEM ACCEPTANCE

1. Make sure you have enough paper in the printer.
2. Login : GYROSCAN.
3. Select: SCAN CONTROL
4. Select: SCAN UTILITIES.
5. Select: ENTER SERVICE MODE
6. Select: SYSTEM TUNING
7. Select: TUNED HARDWARE PARAMETERS
8. Select: LIST.
   A File is created in directory GYRO$TEMP.
9. Select: PRINT
   The file is printed.
10. Add the print-out of the TUNED HARDWARE PARAMETERS to the MRL.

8. SAFETY ACTIONS

8.1. INTRODUCTION

For safety reasons several functions must be tested. Write down the test results in the MRL (section 22).

1. The scan must stop if one of the stop buttons is pressed (on the console or in the RF-room)
2. If the RF room door is opened, the scan must stop.
3. The table movement must stop if the finger protection plate is pressed.
4. A warning message must be displayed if the Helium overpressure is too low.
5. A warning message must be displayed if the Airflow is too low.
6. A warning message must be displayed if the temperature of the gradient coil is too high.

8.2. PROCEDURE FOR SAFETY CHECKS

Preparation:
1. Position the 3 litre bottle lengthwise on the tabletop
2. Move the bottle to the isocentre using the "Travel to scan-plane" button.

Stop scan function, with the console <STOP> function:

1. Login : GYROSCAN
2. Select: New Examination
3. Create Dummy examination
4. Select: < ICON person with slices >.
5. Select: ANATOMY.
6. Select: PHANTOM STUDIES.
7. Select: PIQT.
8. Select: QA3:B,2D,SE
9. Select: <Proceed>
10. Select: START SCAN

After the preparation phase the scanner starts its sequence. Select after a while the <STOP SCAN> function. The scan must stop.

Stop scan function, with the key-pad <STOP> button at the front of the magnet.

1. Select: Scan Control
2. Select: Scan list
3. Select: Planscan
4. Select: Dynamic/ang
5. Set : MANUAL START to yes.
6. Select <Proceed>
7. Select: Start Scan
8. Select: Modified scan

The preparation phase starts.
Wait for the message: Select <Proceed> or press Start Scan at the magnet to (re)start scan

9. Go inside the RF room, close the door, and start the scan with the <Start Scan> button on the PICU. Stop the scan after a while by pressing the <Stop Scan> button on the PICU. The scan must stop.
10. Select. Start Scan
11. Select: <Proceed> to select the stopped scan

The preparation phase starts.

12. Wait for the message: Select <Proceed> or press Start Scan at the magnet to (re)start scan
13. Go inside the RF room, close the door, and start the scan with the <Start Scan> button on the rear side of the magnet. Stop after a while the scan by pressing the <STOP> button on the rear side of the magnet.
14. The scan must stop.

Stop scan function, by opening the RF door:

1. Start a scan from the console and when scanning, open the RF door, the scan must stop. (Connector FMX10 in interface unit NTDAC)

The table movement must stop, if the finger protection plate is pressed:

1. Move the table outwards by pressing the patient table movement knob. Activate the finger protection plate.
2. The table movement must stop. The motor of the hydraulic system must stop immediately.
3. Move the table downwards by pressing the same knob.
4. During the downwards movement press against the finger protection plate and check that the table movement stops.

Check the tabletop release button:

The tabletop release buttons are located on the patient support left and right side.
Check that the tabletop can be moved freely without friction after activating the left tabletop release button.
Repeat this for the right tabletop release button.

Check the tabletop MANUAL function:

Check that the tabletop of the patient support can be moved horizontally without friction after activating the MANUAL switch at the PICU.

Test: Helium overpressure detection circuit:

1. Select a new examination or select another patient. Helium overpressure detection is done only once per examination.
2. Remove one of the connector pins of the helium overpressure sensor (or remove connector MDI5X2 on the DFEC2 unit)
3. Start a scan from the console.

The following warning message must appear on the console: "Helium overpressure too LOW! Contact Philips Service".

4. Re-install the connector pin or re-install the connector MDI5X2.
5. Select another patient and start a scan.
6. No warning message should appear.
7. Stop the scan.

Test: Airflow circuit:

1. Remove connector MDI5X1 on the DFEC2 unit (pressure sensor on back of magnet)
2. Start a scan from the console.
3. When the scan stops, press <Proceed>
4. The following warning message must appear on the console:
   "Gradient Airflow is insufficient. Scan aborted after error.
5. Re-install connector MDI5X1.

Gradient coil temperature sensors detection circuit:

1. Login : GYROSCAN
2. Select: DUMMY EXAMINATION.
3. Select: < ICON person with slices >.
4. Select: ·ANATOMY.
5. Select: PHANTOM STUDIES.
6. Select: PIQT.
7. Select: QA3:B,2D,SE
8. Select: START SCAN.
9. When scanning, disconnect in the gradient power supply interface connector DIBX53.
   The gradient must switch to low power.
10. Press <Proceed>
11. Check if the following warning message appears on the console:
    " Gradient coil is overheated! Wait 15 minutes before restarting." "Scan aborted after error".
12. Reconnect connector DIBX53 in the gradient power supply interface.

Check heat exchanger interlock (only for Intera Power or Master)

1. Start a scan
2. Remove the low water level connector DIBX57 at the gradient amplifier (for systems without water cooling this is a dummy connector).
3. Check that the gradient amplifier switched to low power mode.
4. Press <Proceed>
5. Check for the error message " Low water level detected by the heat exchanger"
6. Mount back connector DIBX57
7. Start a second scan

CAUTION

Do not bend "Coil Return" hose. This can cause high pressure over the gradient coil.

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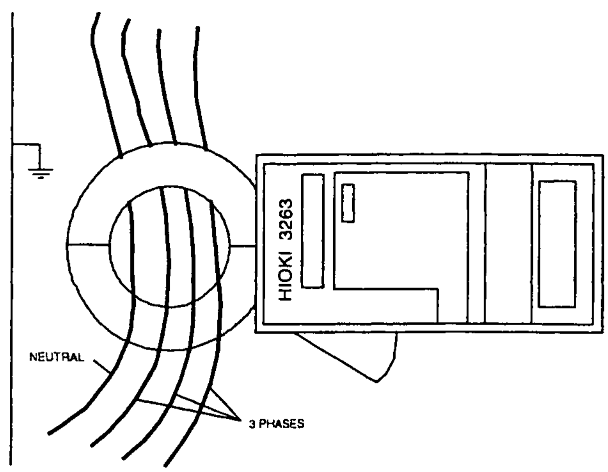
8. Bend the "Coil FEED" hose at the heat exchanger, so that the flow is blocked.
9. The scan should abort within a few seconds.
10. Check that the gradient amplifier switched to low power mode.
11. Check for the error message "Low water flow detected by the heat exchanger".
12. Restore to the original situation.

8.3. LEAKAGE CURRENT MEASUREMENTS

Measure with a HIOKI clamp-on leak tester model 3263-01 the earth leakage current of the system at the input connection of the MDU. Measure the differential current from the "three phases and neutral" See Figure.



Requirements: leakage current (L1,L2,L3,N) < 10 mA

Write the measured values into the MRL.

Measure the leakage current in the following system modes.

- system standby (no activity in TSW or ASW).
- acquisition (during PIQT test).

If the leakage current (L1,L2,L3,N) is more than 10 mA, locate and repair the unit which has a high individual leakage current (>10 mA).

8.4. HELIUM LEVEL CHECK

Magnet monitoring unit

1. Check if the power of the MMU is on. (green led must be on)
2. Press and hold the helium read button for thirty seconds.
3. The 'probe active' led will light
4. Read the HLI and write the value.

Remote Magnet Monitoring unit (RMMU):

1. Login : GYROSCAN.
2. Select: SCAN CONTROL.
3. Select: SCAN UTILITIES.
4. Select: READ HELIUM LEVEL.
   Helium level is ...%